

## KLYUCHI KAMCHATKA, Russian Federation

WMO: 323890

Lat: 56.317N

Lon: 160.833E

Elev: 95

StdP: 14.65

Time Zone: 12.00 (E12)

Period: 94-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |       | Humidification DP/MCDB and HR |     |       |       |     |       | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|-------|-------------------------------|-----|-------|-------|-----|-------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |       | 99.6%                         |     |       | 99%   |     |       | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%   | DP                            | HR  | MCDB  | DP    | HR  | MCDB  | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)   | (d)                           | (e) | (f)   | (g)   | (h) | (i)   | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | -27.1      | -22.8 | -31.3                         | 0.9 | -26.9 | -27.1 | 1.2 | -22.7 | 23.4                  | 13.7 | 21.1 | 15.4 | 2.0                   | 110  | 0.682 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 15.8                   | 77.5            | 64.6 | 74.4 | 62.9 | 71.1 | 61.3 | 66.4                | 74.7 | 64.3 | 72.4 | 62.2 | 69.5 | 5.2                  | 90   |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 63.0                            | 86.4 | 71.0 | 60.7 | 79.5 | 68.9 | 58.7 | 74.1 | 66.5 | 31.1          | 74.7 | 29.5 | 72.4 | 28.0 | 69.6 | 76.1              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |       |                    |     | n-Year Return Period Values of Extreme Temperature |       |            |       |            |       |            |       |      |     |
|-------------------|------|------|----------------------------|-------|--------------------|-----|----------------------------------------------------|-------|------------|-------|------------|-------|------------|-------|------|-----|
|                   |      |      | Mean                       |       | Standard Deviation |     | n=5 years                                          |       | n=10 years |       | n=20 years |       | n=50 years |       |      |     |
| 1%                | 2.5% | 5%   | Min                        | Max   | Min                | Max | Min                                                | Max   | Min        | Max   | Min        | Max   | Min        | Max   |      |     |
| (a)               | (b)  | (c)  | (d)                        | (e)   | (f)                | (g) | (h)                                                | (i)   | (j)        | (k)   | (l)        | (m)   | (n)        | (o)   | (p)  | (q) |
| (4) 23.3          | 20.1 | 17.4 | DB                         | -31.5 | 81.8               | 5.9 | 2.1                                                | -35.7 | 83.3       | -39.2 | 84.5       | -42.5 | 85.7       | -46.8 | 87.2 | (4) |
| (5)               |      |      | WB                         | -31.5 | 69.6               | 6.0 | 2.8                                                | -35.8 | 71.7       | -39.3 | 73.3       | -42.7 | 74.9       | -47.0 | 77.0 | (5) |

## Monthly Climatic Design Conditions

|                                                                                  |                                                                                  | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |      |
|----------------------------------------------------------------------------------|----------------------------------------------------------------------------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|                                                                                  |                                                                                  | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |      |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg                                                                            | 33.0   | 3.9   | 7.8   | 18.2  | 29.4  | 42.5  | 54.6  | 61.2  | 59.1  | 49.7  | 36.8  | 22.0  | 8.8   |      |
|                                                                                  | DBStd                                                                            | 22.10  | 17.09 | 13.26 | 10.45 | 6.75  | 6.13  | 6.50  | 5.49  | 4.63  | 5.39  | 6.63  | 9.63  | 14.12 |      |
|                                                                                  | HDD50                                                                            | 7084   | 1429  | 1182  | 985   | 617   | 248   | 27    | 1     | 0     | 69    | 410   | 840   | 1276  |      |
|                                                                                  | HDD65                                                                            | 11727  | 1894  | 1602  | 1450  | 1067  | 698   | 317   | 143   | 192   | 458   | 875   | 1290  | 1741  |      |
|                                                                                  | CDD50                                                                            | 871    | 0     | 0     | 0     | 0     | 15    | 164   | 348   | 283   | 61    | 0     | 0     | 0     |      |
|                                                                                  | CDD65                                                                            | 39     | 0     | 0     | 0     | 0     | 0     | 5     | 25    | 9     | 0     | 0     | 0     | 0     |      |
|                                                                                  | CDH74                                                                            | 289    | 0     | 0     | 0     | 0     | 1     | 54    | 189   | 45    | 0     | 0     | 0     | 0     |      |
|                                                                                  | CDH80                                                                            | 14     | 0     | 0     | 0     | 0     | 0     | 3     | 10    | 1     | 0     | 0     | 0     | 0     |      |
| Wind                                                                             | WSAvg                                                                            | 6.8    | 6.0   | 7.4   | 7.7   | 7.6   | 6.5   | 6.0   | 5.1   | 5.3   | 6.6   | 7.4   | 8.5   | 7.1   |      |
| Precipitation                                                                    | PrecAvg                                                                          | 24.3   | 2.7   | 2.3   | 1.7   | 1.2   | 1.5   | 1.6   | 1.9   | 2.7   | 2.0   | 2.1   | 2.4   | 2.4   |      |
|                                                                                  | PrecMax                                                                          | 34.7   | 5.7   | 4.7   | 4.4   | 3.4   | 3.4   | 4.7   | 3.9   | 7.3   | 6.9   | 5.7   | 4.8   | 4.8   |      |
|                                                                                  | PrecMin                                                                          | 18.0   | 0.4   | 0.6   | 0.0   | 0.1   | 0.3   | 0.1   | 0.2   | 0.7   | 0.1   | 0.0   | 0.3   | 0.8   |      |
|                                                                                  | PrecStd                                                                          | 4.5    | 1.3   | 1.3   | 1.0   | 0.7   | 0.7   | 1.1   | 0.9   | 1.3   | 1.1   | 1.3   | 1.3   | 1.0   |      |
| Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%                                                                             | DB     | 34.5  | 34.8  | 36.8  | 48.5  | 70.1  | 78.9  | 81.4  | 79.0  | 67.5  | 52.9  | 41.6  | 37.4  |      |
|                                                                                  |                                                                                  | MCWB   | 33.0  | 32.7  | 32.6  | 40.1  | 54.9  | 61.6  | 67.4  | 68.3  | 58.4  | 44.7  | 37.7  | 35.7  |      |
|                                                                                  | 2%                                                                               | DB     | 33.0  | 32.7  | 35.2  | 43.8  | 63.4  | 75.1  | 78.6  | 74.3  | 63.9  | 50.1  | 38.7  | 33.2  |      |
|                                                                                  |                                                                                  | MCWB   | 31.9  | 31.3  | 31.9  | 36.3  | 50.7  | 60.4  | 65.9  | 64.4  | 55.9  | 43.2  | 35.6  | 31.8  |      |
|                                                                                  | 5%                                                                               | DB     | 31.2  | 30.9  | 33.7  | 40.4  | 59.1  | 71.3  | 75.9  | 70.9  | 61.0  | 47.8  | 36.3  | 31.5  |      |
|                                                                                  |                                                                                  | MCWB   | 30.2  | 29.7  | 31.1  | 34.1  | 48.0  | 58.5  | 64.4  | 62.2  | 54.0  | 42.4  | 33.5  | 30.7  |      |
|                                                                                  | 10%                                                                              | DB     | 28.0  | 27.5  | 32.2  | 37.9  | 54.6  | 67.3  | 72.9  | 67.9  | 58.2  | 45.8  | 34.1  | 28.7  |      |
|                                                                                  |                                                                                  | MCWB   | 27.1  | 26.1  | 30.1  | 32.8  | 45.4  | 56.3  | 63.0  | 60.2  | 52.3  | 41.3  | 31.8  | 27.5  |      |
|                                                                                  | Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%   | WB    | 33.7  | 33.3  | 34.2  | 40.6  | 55.1  | 65.0  | 69.8  | 69.4  | 60.0  | 47.3  | 38.9  | 35.9 |
|                                                                                  |                                                                                  |        | MCDB  | 34.2  | 34.3  | 35.3  | 47.4  | 68.8  | 74.8  | 78.8  | 77.1  | 65.9  | 50.1  | 40.6  | 37.4 |
| 2%                                                                               |                                                                                  | WB     | 32.1  | 31.6  | 32.8  | 36.8  | 51.1  | 61.4  | 67.2  | 65.8  | 56.9  | 45.0  | 36.4  | 32.3  |      |
|                                                                                  |                                                                                  | MCDB   | 32.8  | 32.5  | 34.2  | 42.4  | 62.8  | 72.1  | 75.9  | 72.7  | 61.5  | 48.0  | 37.9  | 32.8  |      |
| 5%                                                                               |                                                                                  | WB     | 30.2  | 29.8  | 31.7  | 34.7  | 48.3  | 59.3  | 65.3  | 63.2  | 54.9  | 43.5  | 34.3  | 30.4  |      |
|                                                                                  |                                                                                  | MCDB   | 31.1  | 30.9  | 33.4  | 38.9  | 58.3  | 69.7  | 74.0  | 69.4  | 59.1  | 46.7  | 35.7  | 31.3  |      |
| 10%                                                                              |                                                                                  | WB     | 27.1  | 26.1  | 30.0  | 33.4  | 45.7  | 57.0  | 63.6  | 61.1  | 53.2  | 41.8  | 32.3  | 27.6  |      |
|                                                                                  |                                                                                  | MCDB   | 28.1  | 27.5  | 32.1  | 36.9  | 53.9  | 66.4  | 71.9  | 66.6  | 56.9  | 45.1  | 33.8  | 28.8  |      |
| Mean Daily<br>Temperature<br>Range                                               |                                                                                  | 5% DB  | MDBR  | 11.7  | 13.4  | 14.6  | 12.7  | 15.1  | 16.3  | 15.8  | 14.0  | 13.3  | 10.3  | 9.9   | 11.2 |
|                                                                                  |                                                                                  |        | MCDBR | 9.5   | 9.1   | 10.2  | 15.3  | 24.2  | 24.9  | 23.0  | 19.7  | 17.3  | 12.9  | 9.5   | 10.8 |
|                                                                                  | MCWBR                                                                            |        | 9.0   | 8.7   | 7.6   | 10.1  | 14.0  | 12.8  | 12.1  | 11.0  | 10.1  | 8.5   | 7.7   | 10.3  |      |
|                                                                                  | 5% WB                                                                            | MCDBR  | 8.9   | 9.1   | 8.4   | 12.9  | 23.2  | 22.2  | 20.7  | 18.0  | 14.7  | 10.7  | 8.8   | 10.0  |      |
|                                                                                  |                                                                                  | MCWBR  | 9.0   | 8.9   | 6.9   | 8.8   | 13.9  | 12.4  | 11.7  | 10.7  | 9.2   | 7.7   | 7.6   | 9.9   |      |
| Clear-Sky<br>Solar<br>Irradiance                                                 | taub                                                                             | 0.273  | 0.310 | 0.368 | 0.433 | 0.490 | 0.460 | 0.479 | 0.454 | 0.372 | 0.342 | 0.286 | 0.265 |       |      |
|                                                                                  | taud                                                                             | 2.047  | 2.019 | 1.881 | 1.771 | 1.786 | 2.044 | 2.053 | 2.118 | 2.330 | 2.209 | 2.103 | 2.006 |       |      |
|                                                                                  | Ebn at Noon                                                                      | 187    | 227   | 240   | 242   | 237   | 249   | 240   | 237   | 245   | 216   | 180   | 154   |       |      |
|                                                                                  | Edh at Noon                                                                      | 20     | 31    | 46    | 60    | 63    | 49    | 48    | 43    | 30    | 26    | 18    | 16    |       |      |
| All-Sky Solar<br>Radiation                                                       | RadAvg                                                                           | 209    | 477   | 921   | 1436  | 1687  | 1652  | 1618  | 1230  | 922   | 514   | 250   | 141   |       |      |
|                                                                                  | RadStd                                                                           | 23     | 58    | 92    | 119   | 157   | 146   | 146   | 122   | 89    | 44    | 16    | 14    |       |      |

## Historical Trends

|                            | DBAvg | Heating |        | Cooling |       |       | Degree-Days |       |       |       |
|----------------------------|-------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|
|                            |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
|                            |       | (d)     | (e)    | (f)     | (g)   | (h)   | (i)         | (j)   | (k)   | (l)   |
| (46) Station Only          | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |
| (47) Regional (1 neighbor) | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |

Nomenclature: See separate page